Embedded Controller

General Purpose I/O (GPIO)

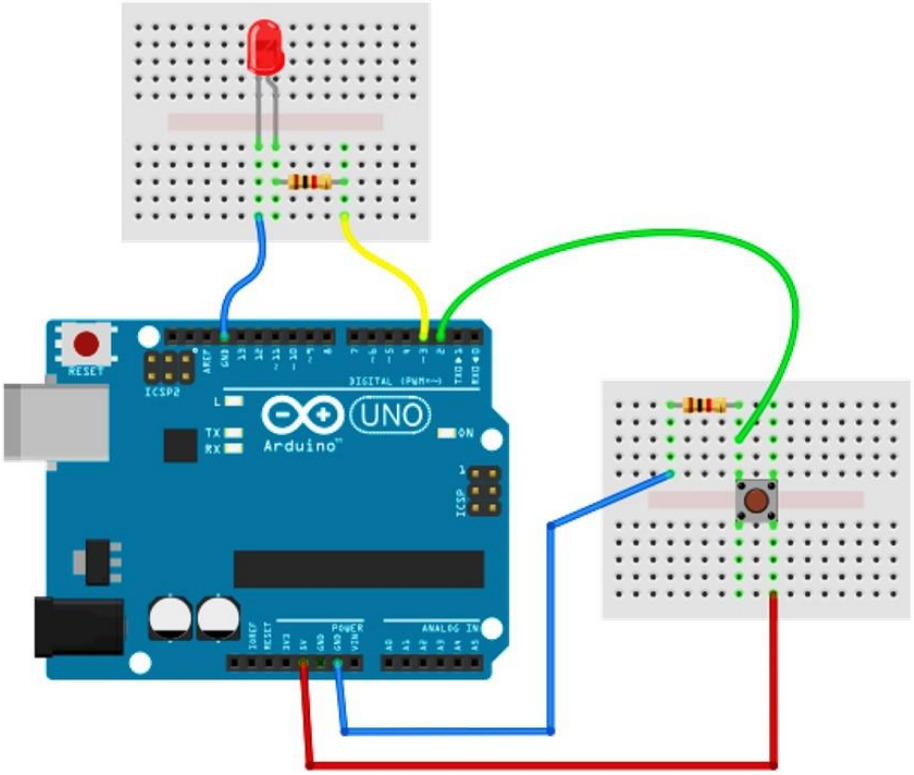
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School of Mechanical & Control Engineering

Last modified: 2025.8

Background

- **MCU is a digital system**
  - output H(L) voltage or read H(L) signal
- **How to turn LED ON/OFF with MCU ?**
  - How to supply High or Low Voltage from MCU to LED?
  - How can we continuously supply Voltage to LED from Digital Pin?

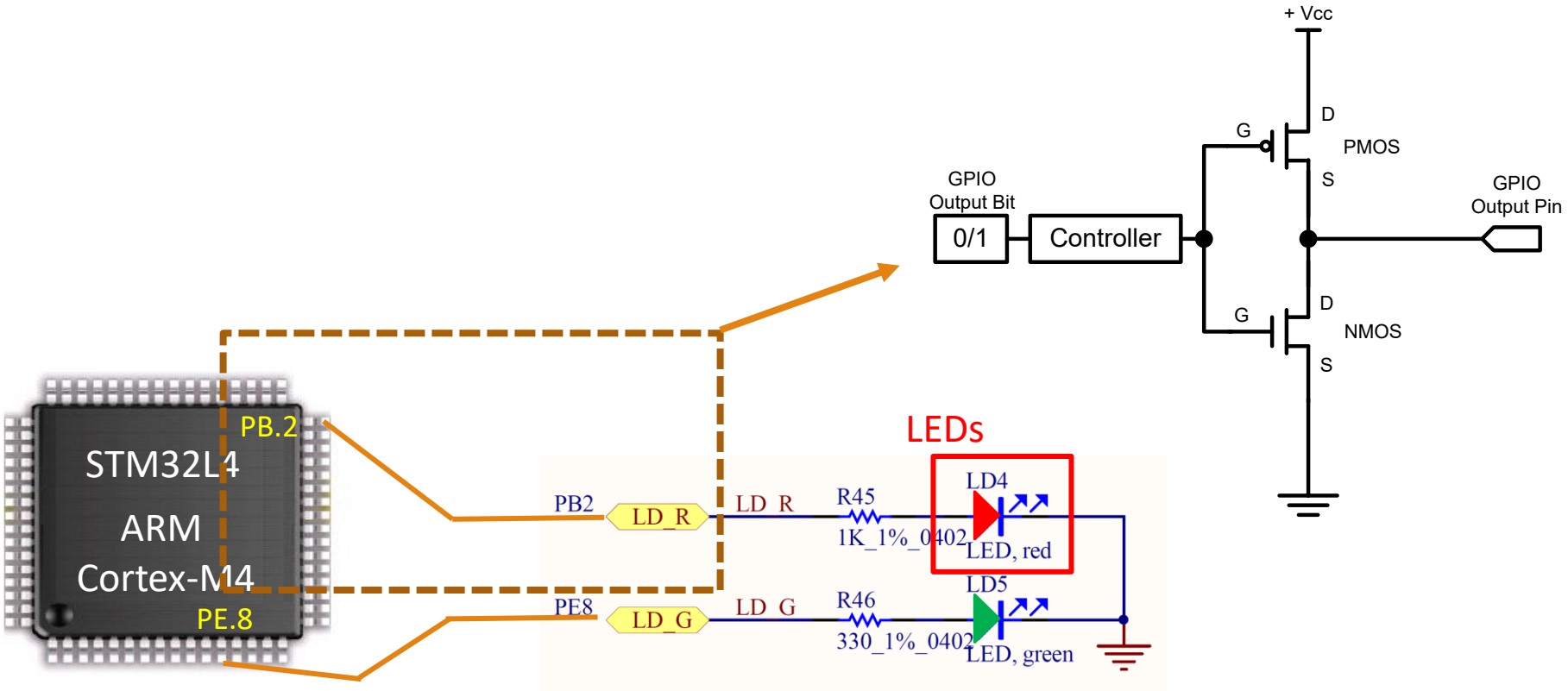
- **How can we use the same Pin for Digital Input or/and Output?**



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What we will learn

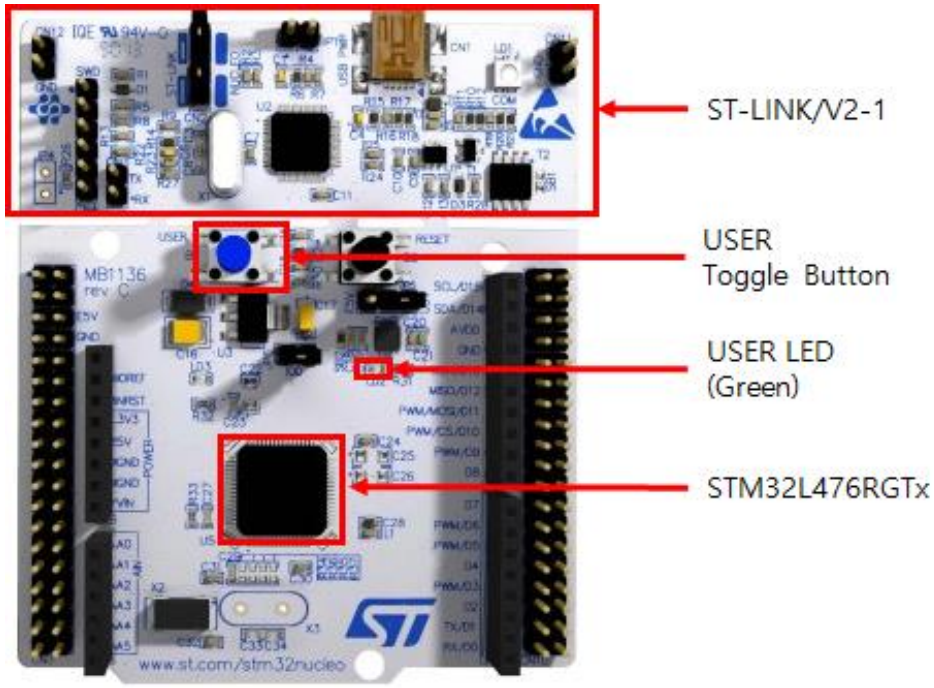
- **Topics**
  - What is GPIO (Genera Purpose I/O)
  - How digital In/Out of GPIO works
  - What is Pull-down/Pull-up
  - How to configure GPIO Digital In/Out



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What is GPIO

- **GPIO: General-purpose I/O**
  - Pins of a processor that can perform multiple functions
  - 32-bit MCU typically has Port A~H (8 ports)
  - Each Port has up to 16 or 32 pins
    - GPIOA[0:15]
    - GPIOA[0:31]

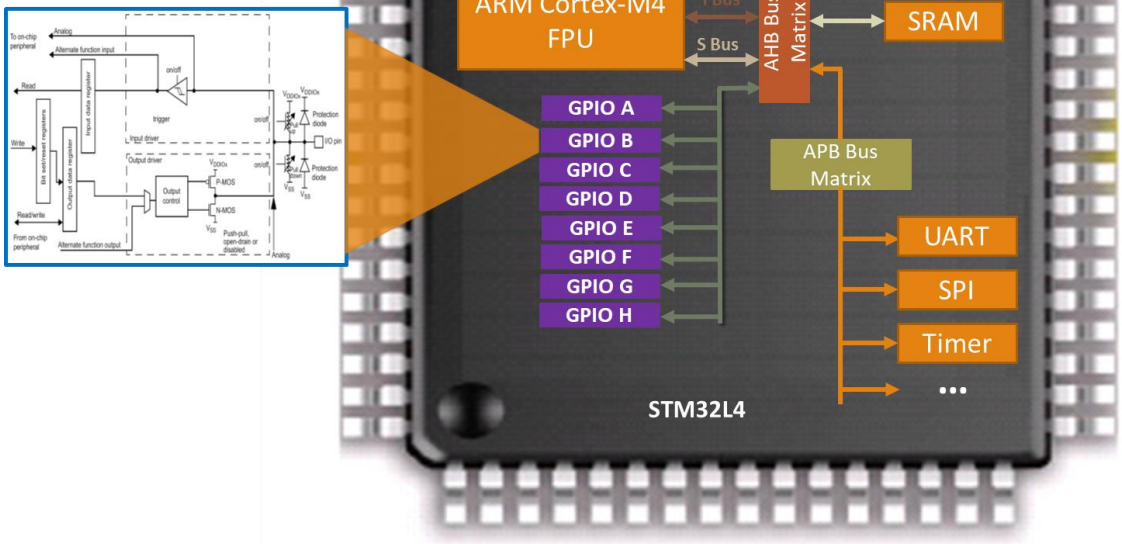


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What is GPIO

- **Example: GPIO of STM32f4**
  - 32-bit Port A~H (8 ports)
  - Each Port has 32 pins: GPIO_A[0:31]
    - But usually use 0~15

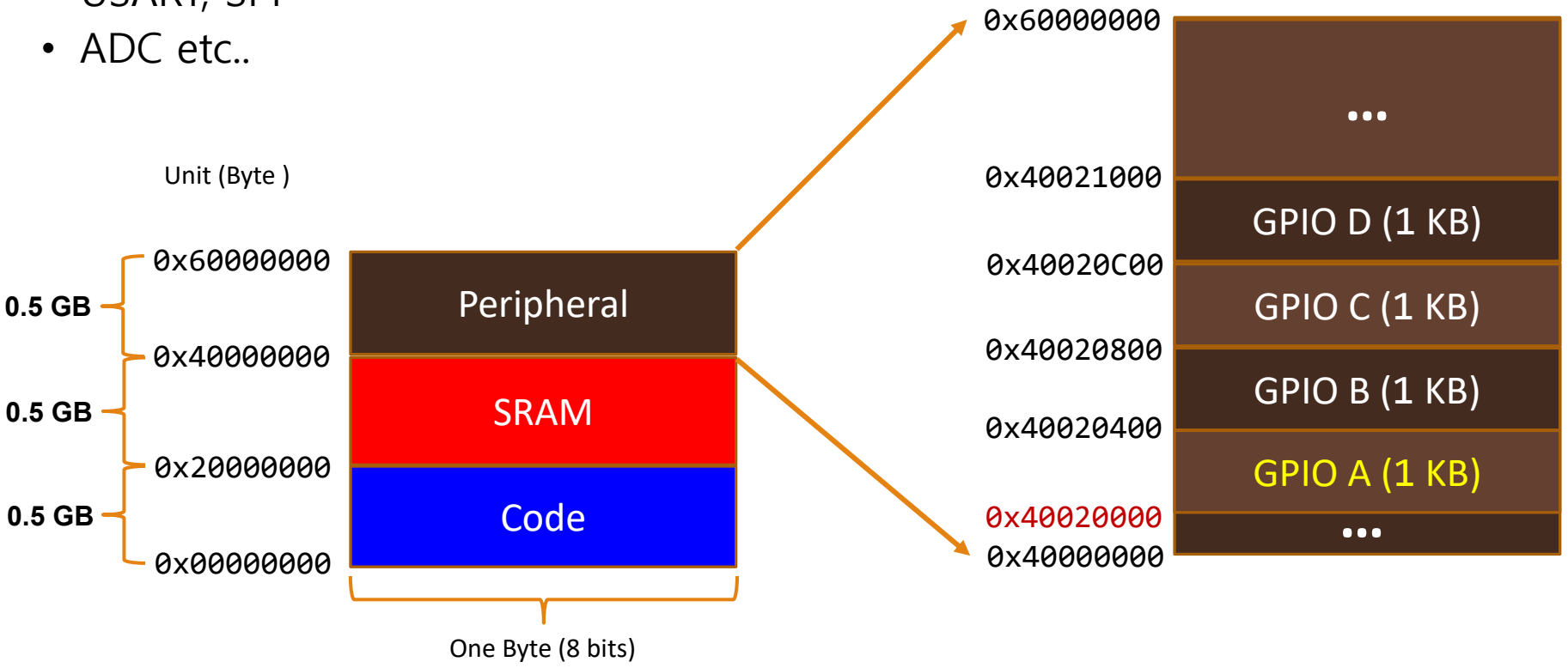
- **Pin mode**
  - Digital Input
  - Digital Out
  - Analog function
  - Alternate functions



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GPIO Register

- **GPIO is memory-mapped I/O**
  - Peripheral Register:
    - GPIO
    - Timers
    - USART, SPI
    - ADC etc..



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GPIO Register

- **For STM32F411**
  - GPIOA starts at 0x4002 0000

<Reference Manual(rev3): pg. 38>

Table 1. STM32F411xC/E register boundary addresses

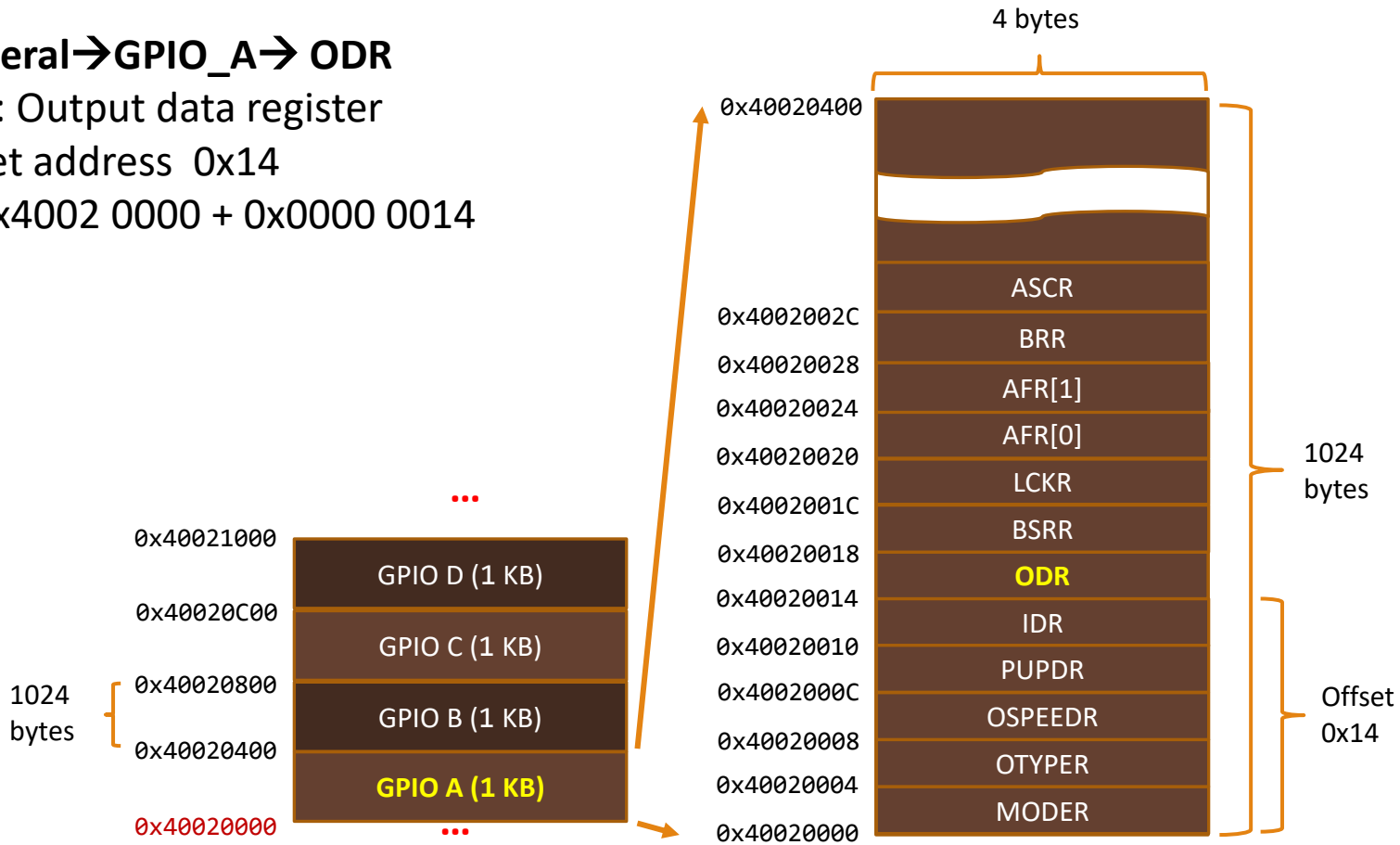
Boundary address	Peripheral	Bus	Register map
0x5000 0000 - 0x5003 FFFF	USB OTG FS	AHB2	*Section 22.16.6: OTG_FS register map on page 754*
0x4002 6400 - 0x4002 67FF	DMA2	AHB1	*Section 9.5.11: DMA register map on page 197*
0x4002 6000 - 0x4002 63FF	DMA1		
0x4002 3C00 - 0x4002 3FFF	Flash interface register		*Section 3.8: Flash interface registers on page 59*
0x4002 3800 - 0x4002 3BFF	RCC		*Section 6.3.22: RCC register map on page 136*
0x4002 3000 - 0x4002 33FF	CRC		*Section 4.4.4: CRC register map on page 69*
0x4002 1C00 - 0x4002 1FFF	GPIOH		*Section 8.4.11: GPIO register map on page 163*
0x4002 1000 - 0x4002 13FF	GPIOE		
0x4002 0C00 - 0x4002 0FFF	GPIOD		
0x4002 0800 - 0x4002 0BFF	GPIOC		
0x4002 0400 - 0x4002 07FF	GPIOB		
0x4002 0000 - 0x4002 03FF	GPIOA		
		*	
		*	
		*	
0x4000 0C00 - 0x4000 0FFF	TIM5		*Section 13.4.21: TIMx register map on page 373*
0x4000 0800 - 0x4000 0BFF	TIM4		
0x4000 0400 - 0x4000 07FF	TIM3		
0x4000 0000 - 0x4000 03FF	TIM2		

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GPIO Register

- **Peripheral→GPIO_A**
  - Port memory size is 1024(0x400) bytes

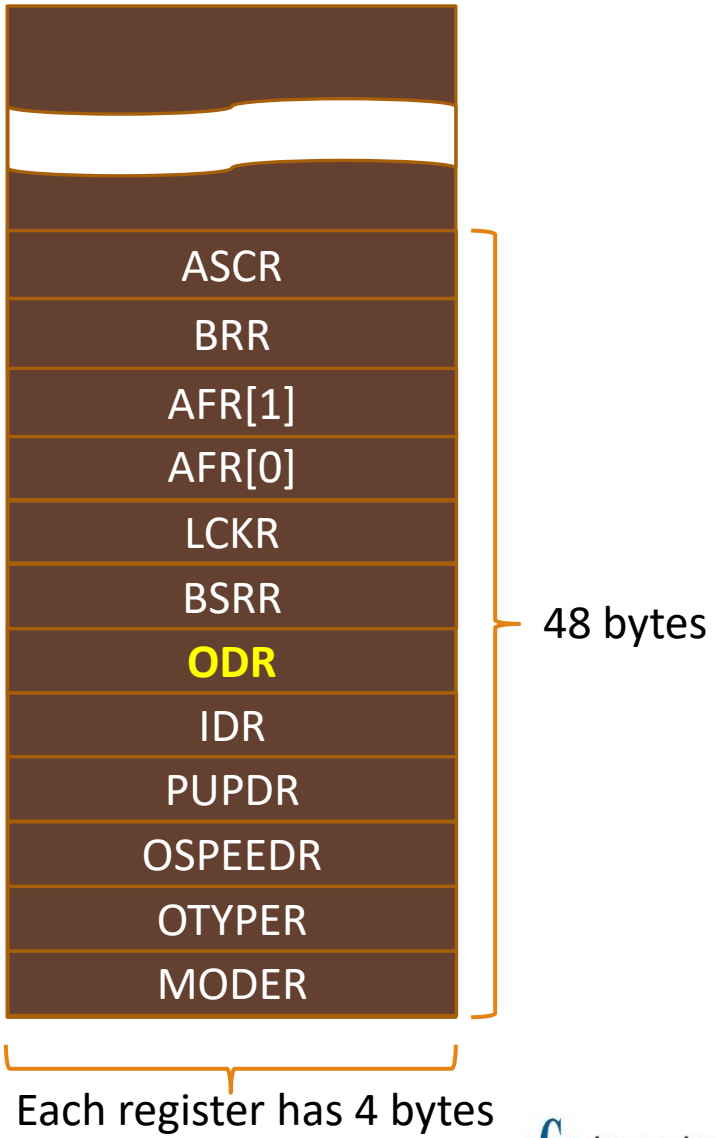
- **Peripheral→GPIO_A→ ODR**
  ODR: Output data register
  Offset address 0x14
  → 0x4002 0000 + 0x0000 0014



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GPIO Register

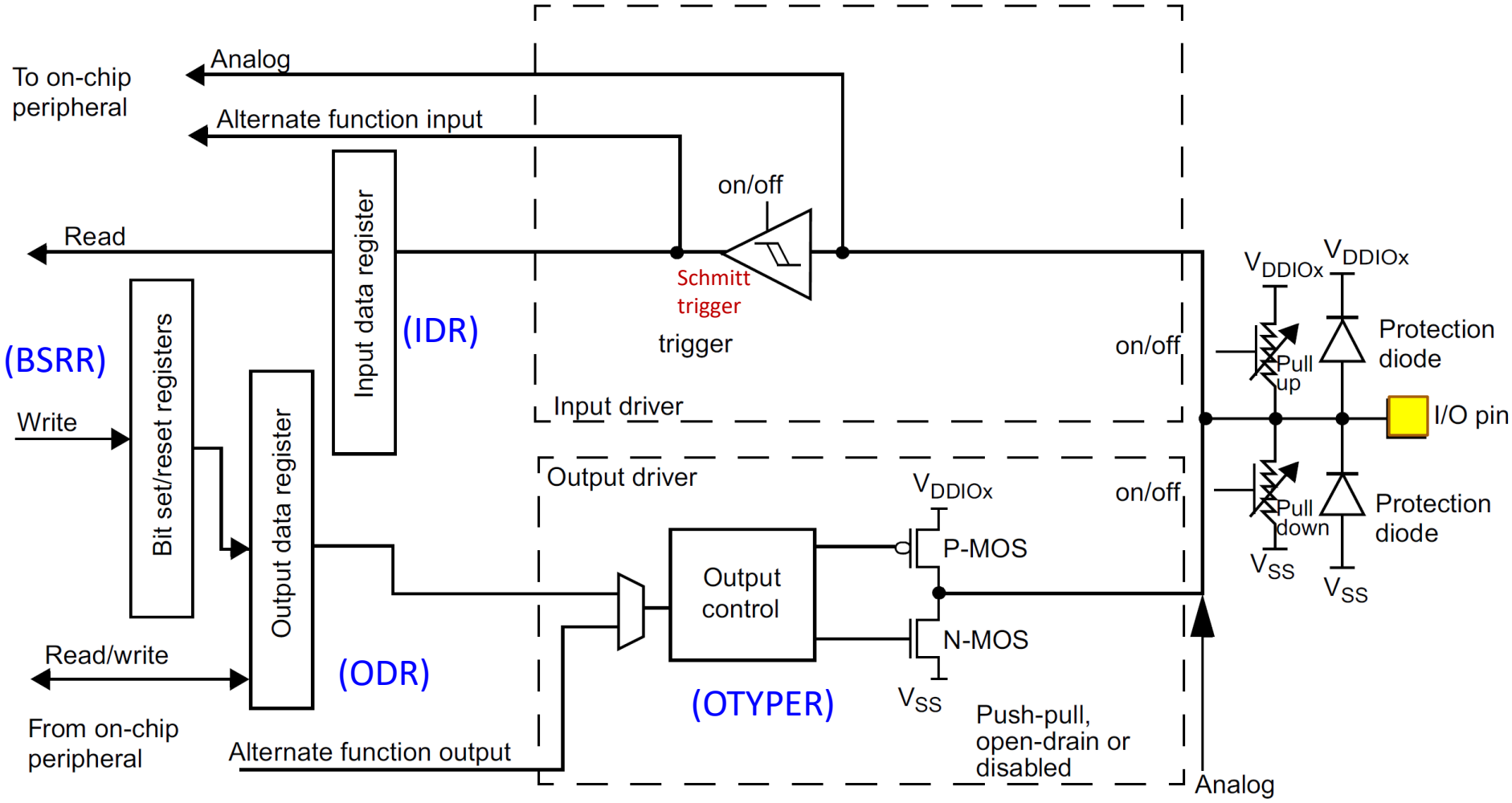
- **GPIO Register components**
  - **Configuration registers (4)**
    - GPIOx_MODER
    - GPIOx_OTYPER
    - GPIOx_OSPEEDR
    - GPIOx_PUPDR
  - **Data registers (2)**
    - GPIOx_IDR
    - GPIOx_ODR
  - **Set/reset register (1)**
    - GPIOx_BSRR
  - **Locking register (1)**
    - GPIOx_LCKR
  - **Alternate function selection register (2)**
    - GPIOx_AFRH
    - GPIOx_AFRL



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GPIO Hardware Circuit

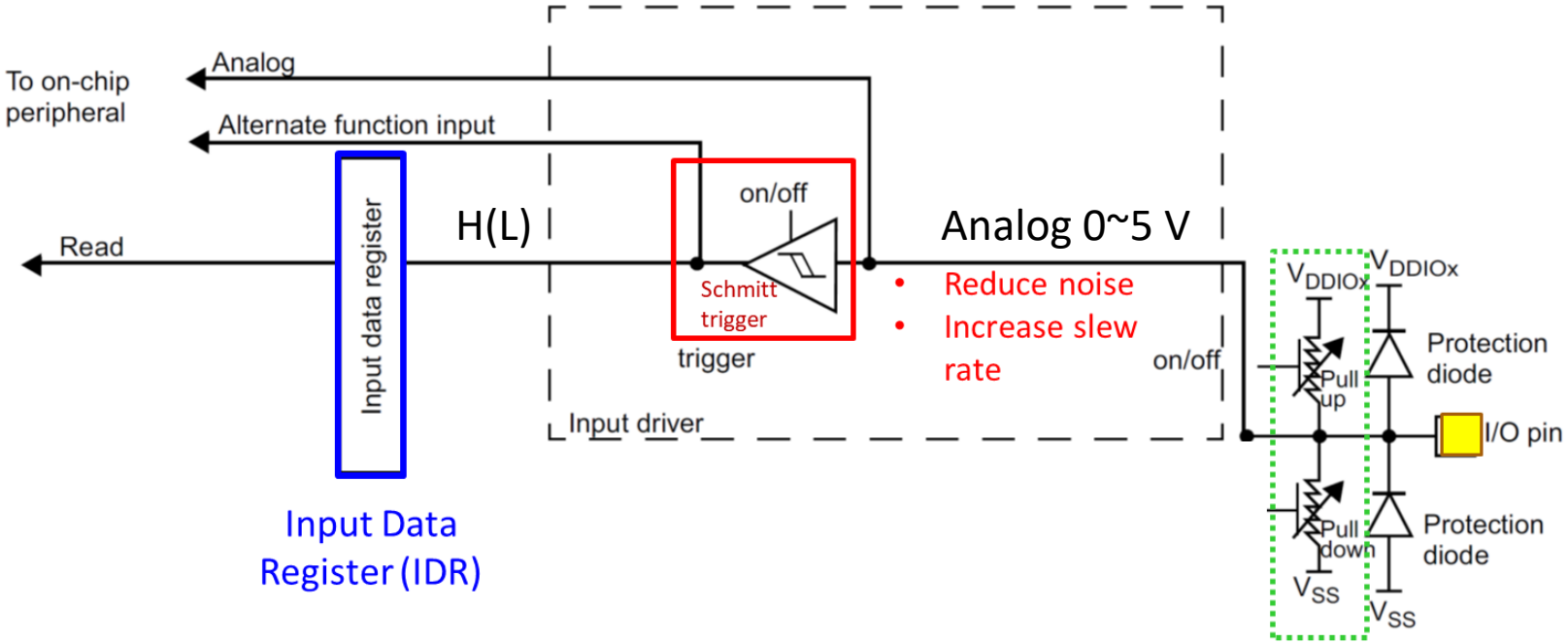
- Digital Out/IN can be controlled with GPIO registers



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GPIO: Digital Input

- **Steps for configuration of GPIO Digital Input**
  1. Configure as Digital Input (**MODE**)
     - Schmitt trigger input is activated
     - Output buffer is disabled
  2. Configure pull-up/down resistors (**PUPD**)
  3. Read Data from Input data register**(ID)**



GPIO Pull-up/Pull-down Register (PUPDR)

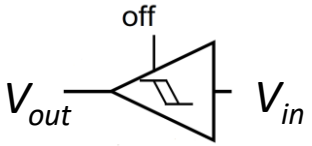
00 = No pull-up, pull-down 01 = Pull-up

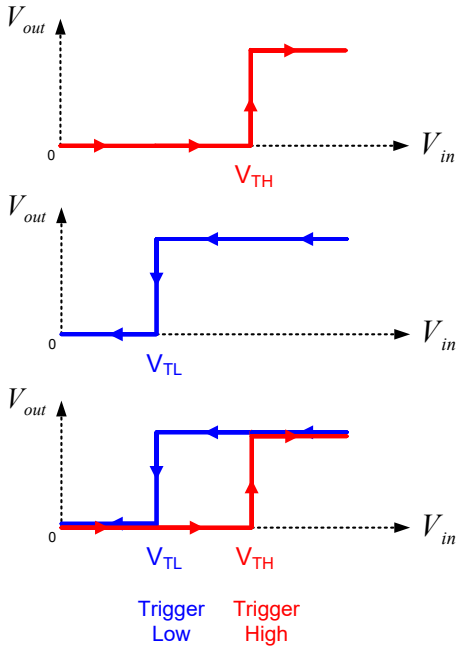
10 = Pull-down 11 = Reserved

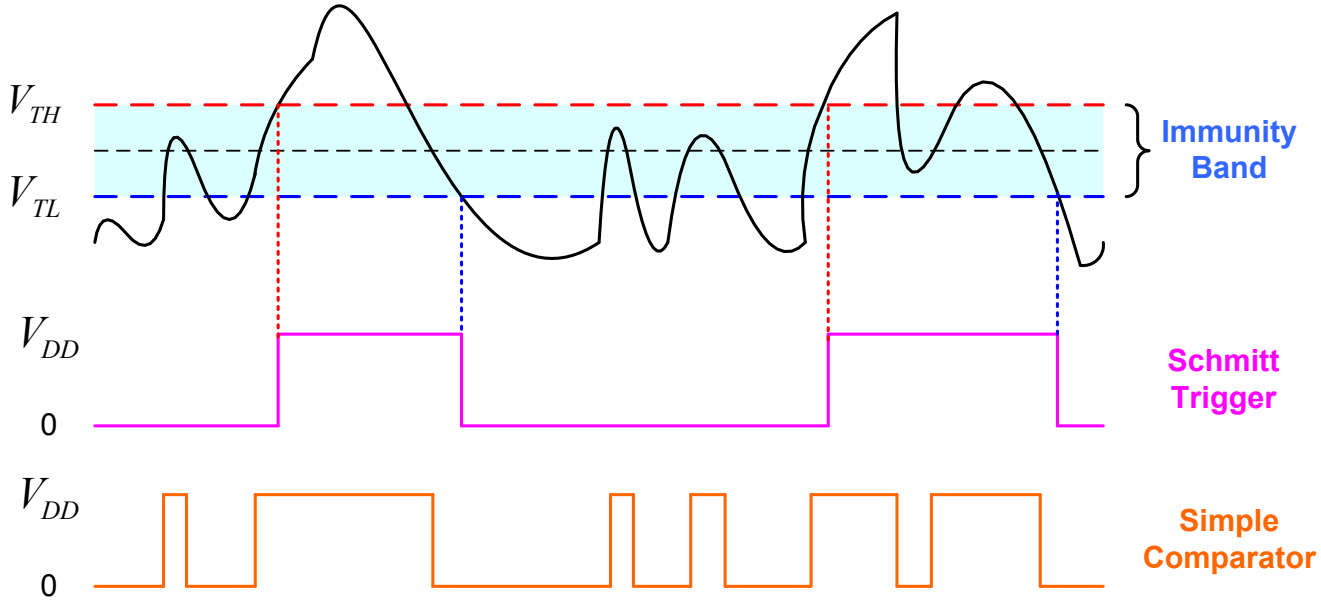
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GPIO: Digital Input

- **Schmitt Trigger:**
- A voltage comparator with hysteresis
- From Analog Input to Outputs H or L signals
- Used to convert noisy or slow signal edge → clean edge signal
- Immune to undesired noise







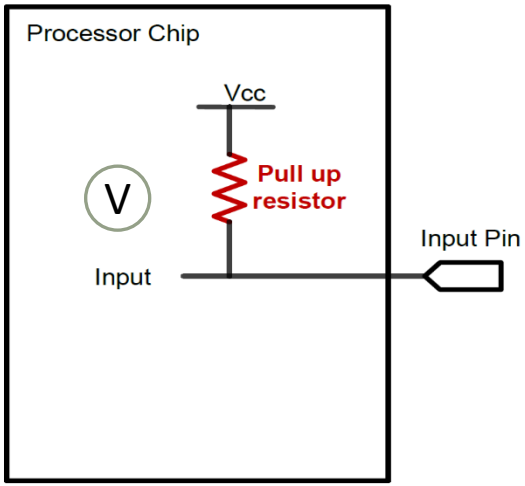
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GPIO: Digital Input

- **Pull-Up / Pull-Down**
  - A digital input can have three states
    - High, Low, and High-Impedance (also called floating, Z)
  - Pull-up/down: To ensure input either low(0) or high(1)

- Pull-up

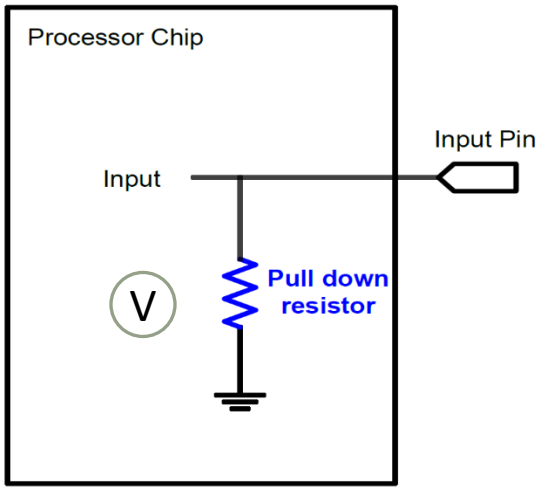
- InputPin=Z → Input=H
- InputPin=L → Input=H
- InputPin=H → Input=L



Pull-Up

- Pull-down

- InputPin=Z → Input=L
- InputPin=L → Input=L
- InputPin=H → Input=H

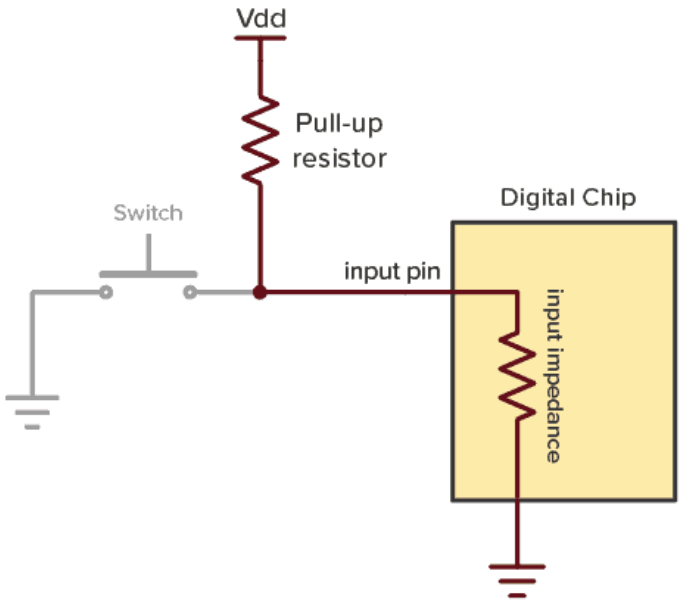


Pull-Down

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GPIO: Digital Input

- **Pull-Up / Pull-Down**
  - Use Pull-Up or Pull-Down for Digital Input
    - switch input, digital sensor input to avoid floating state
  - Pull-Up/Pull-down resistor:
    - at least 10 times smaller than the input impedance of the pin
    - typical pull-up resistor value is 1-10 kΩ.



GPIO: Digital Input

- **Strong vs Weak pull-up/down***
  - Weak pull-up/down: large resistance
  - Strong pull-up/down: small resistance

  - Internal pull-up/down resistors are **large value**, typically tens of kOhm
    - To limit current down to the processor

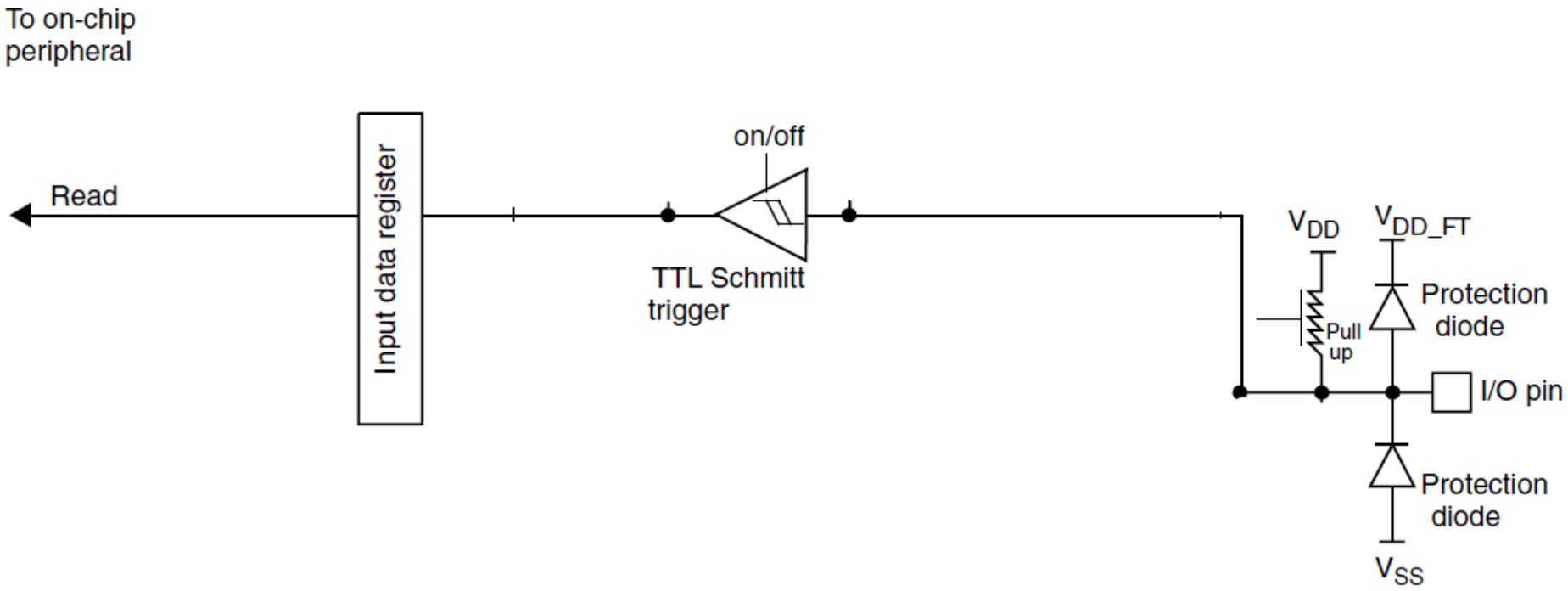
  - If Input Pin is connected externally with a capacitance
    → large resistance & capacitance → large impedance
    → slow in change voltage level
    → Low speed pin/ not accepted for fast communication protocol

  - For faster speed
    → use small resistor (several kOhm) externally

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GPIO: Digital Input

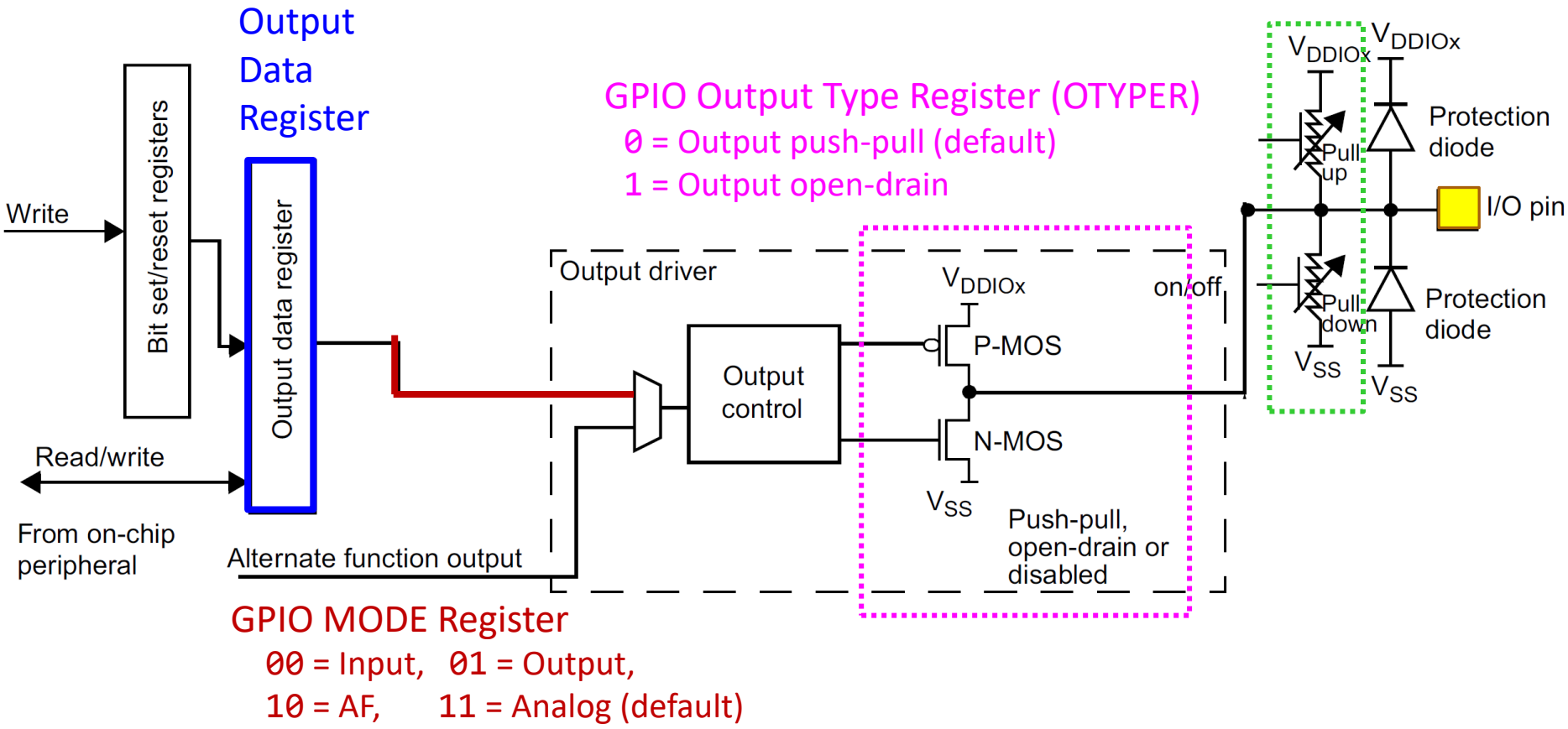
- **Example of Digital Input Configuration**
  - MODER: Input
  - PUPDR: Pull-Up
  - IDR: Read data



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Digital Output: Registers

GPIO Pull-up/Pull-down Register (PUPDR)
00 = No pull-up, pull-down 01 = Pull-up
10 = Pull-down 11 = Reserved



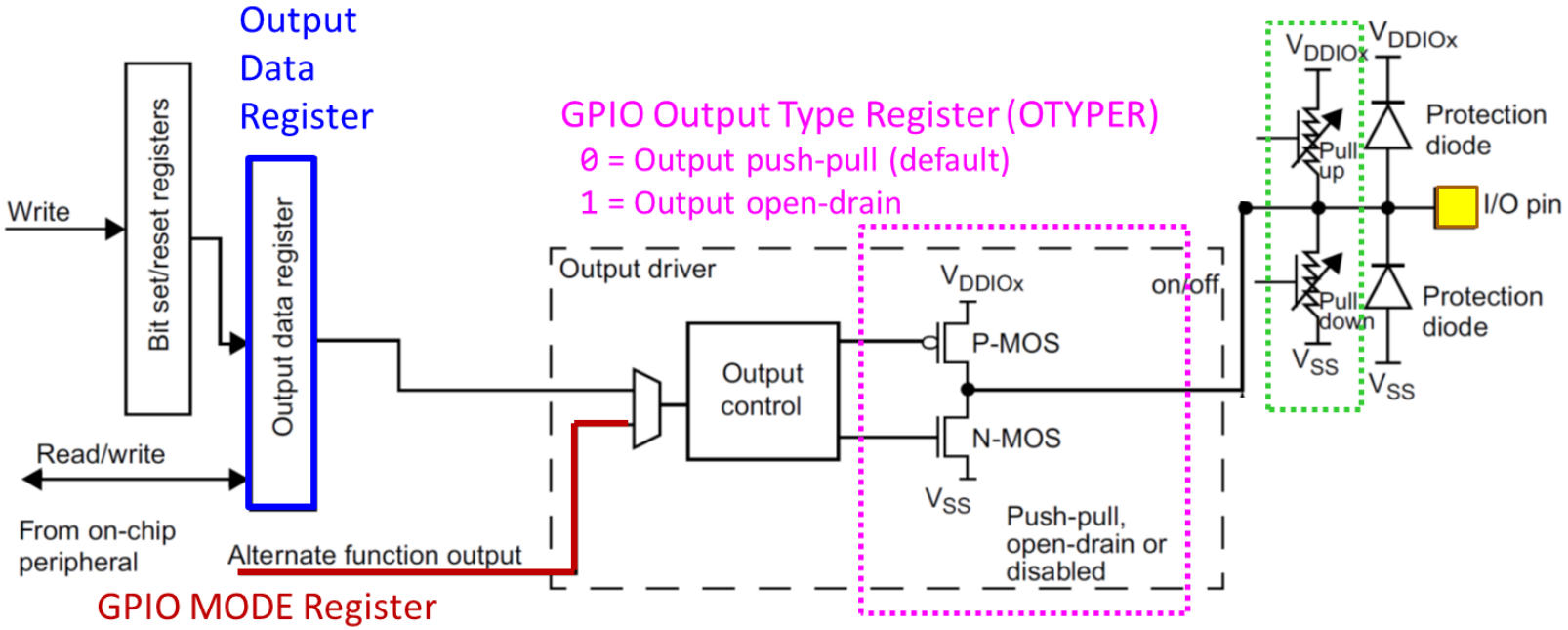
GPIO MODE Register
00 = Input, 01 = Output,
10 = AF, 11 = Analog (default)

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GPIO: Digital Out

- **Steps for configuration of GPIO Digital Output**
  1. Configure as Digital Output (**MODER**)
  2. Configure pull-up/down (**PUPDR**)
  3. Configure output type (**OTYPER**)
     - Open-drain, Push-Pull
  4. Configure output speed **(OSPEER)**
  5. Output Data from ODR to Pin **(ODR)**

GPIO Pull-up/Pull-down Register (PUPDR)
00 = No pull-up, pull-down 01 = Pull-up
10 = Pull-down 11 = Reserved

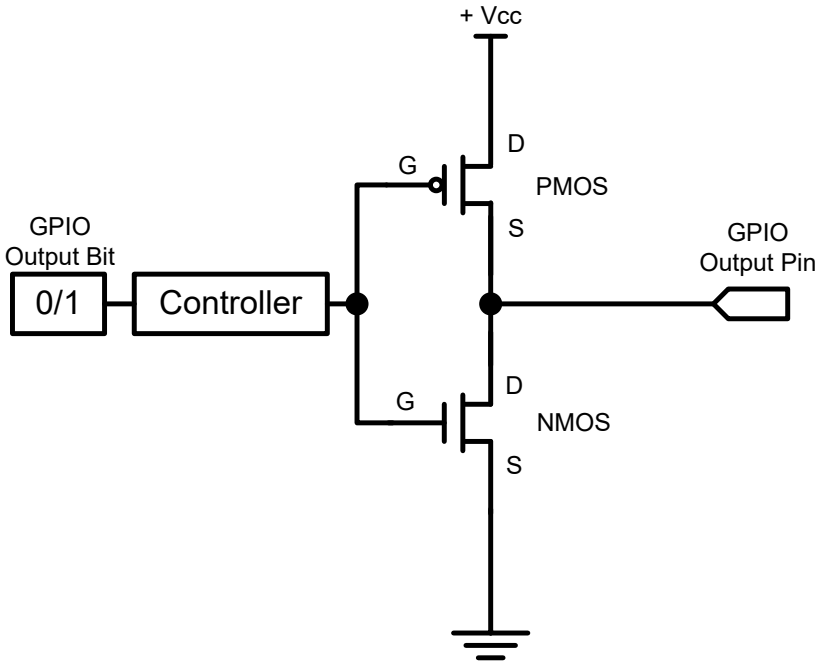


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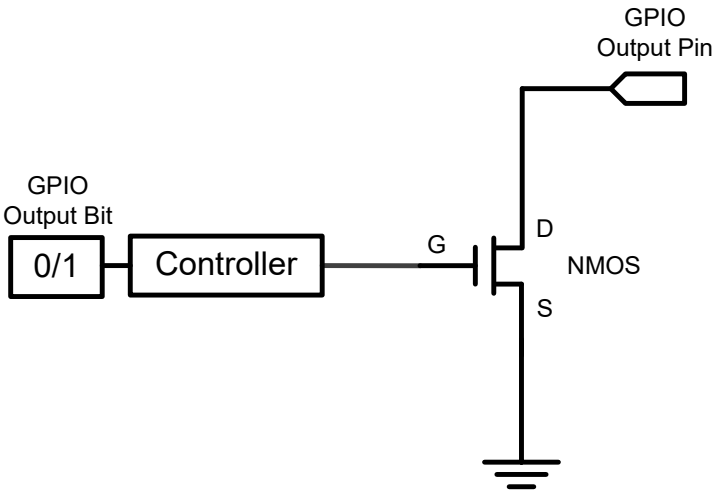
GPIO: Digital Out

- **Push-Pull vs Open Drain**
  - Pull-Pull: Allows the pin to supply and absorb current
  - Open Drain: Allows the pin to only absorb current

- **Push-Pull**
  - A pair of Complementary trans.
  - Only one is turned on at a time



- **Open drain**
  - Drain('0') or Open('Z') states
  - Often has external pull-up

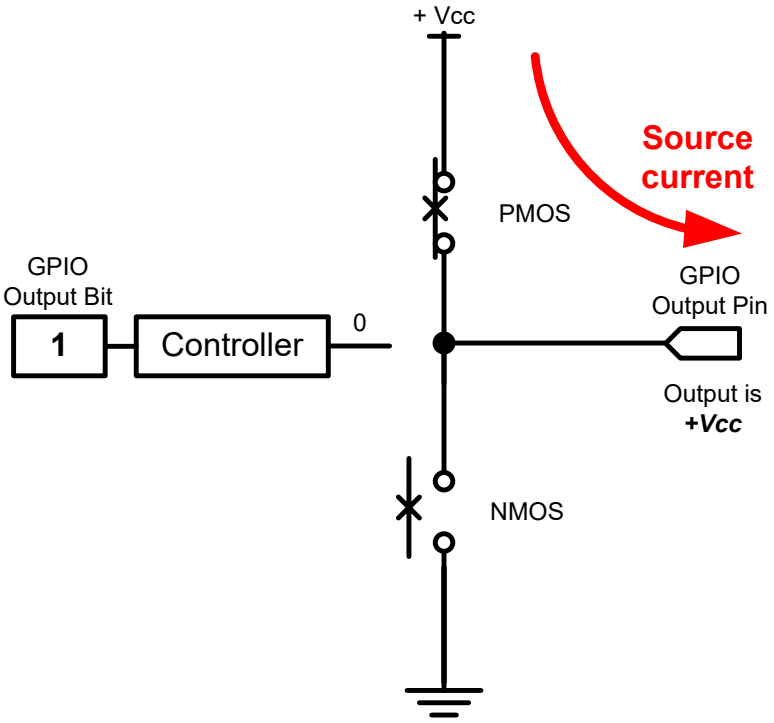


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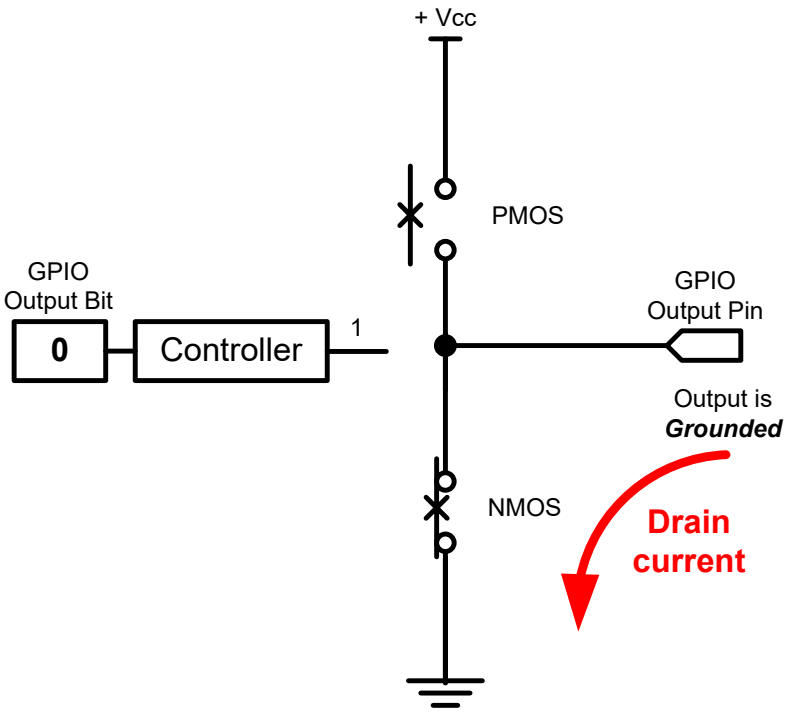
GPIO: Digital Out

- **Push-Pull**
  - Allows the pin to supply and absorb current
  - A pair of complementary transistors.
  - Only one is turned on at a time

GPIO Output = 1
Source current to external circuit



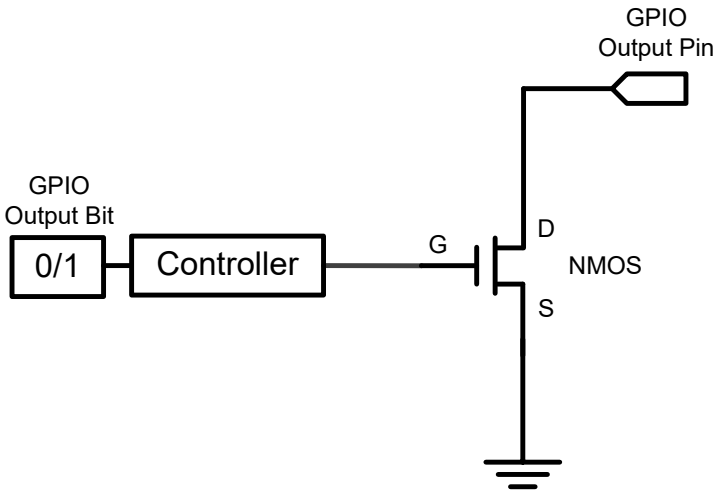
GPIO Output = 0
Drain(sink) current from external circuit



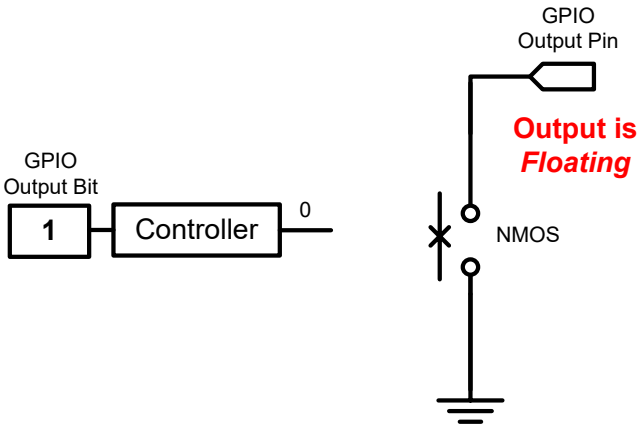
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GPIO: Digital Out

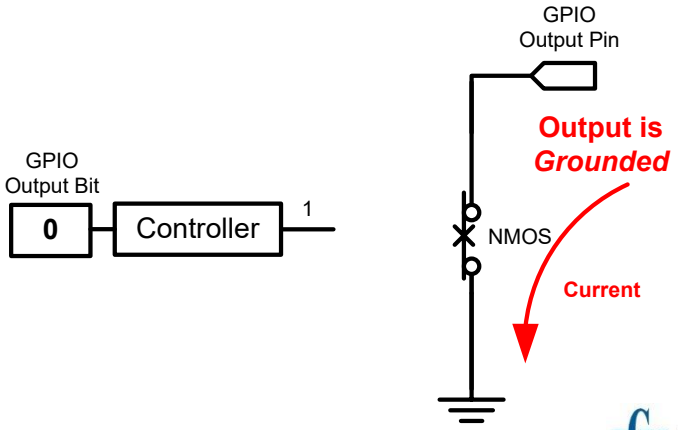
- **Open drain**
  - Allows the pin to only absorb current
  - A pair of *same type* transistors.
  - Drain('0') or Open('Z') states
  - Often has external pull-up



Output = 1
GPIO Pin has high-impedance to external circuit



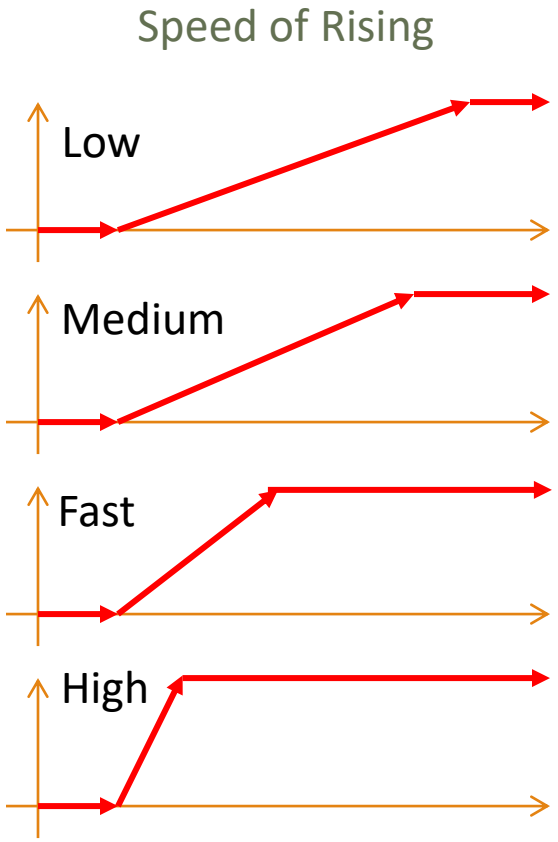
GPIO Output = 0
Drain current from external circuit



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GPIO: Digital Out

- **Output Speed:**
  - Speed of rising and falling
  - Four speeds:
    - Low (~400kHz)
    - Medium (~2MHz)
    - Fast (~10MHz)
    - High (~40MHz)



- **Tradeoff**
  - Higher GPIO speed increases EMI noise and power consumption
  - Configure based on peripheral speed
    - Low speed for toggling LEDs
    - High speed for SPI

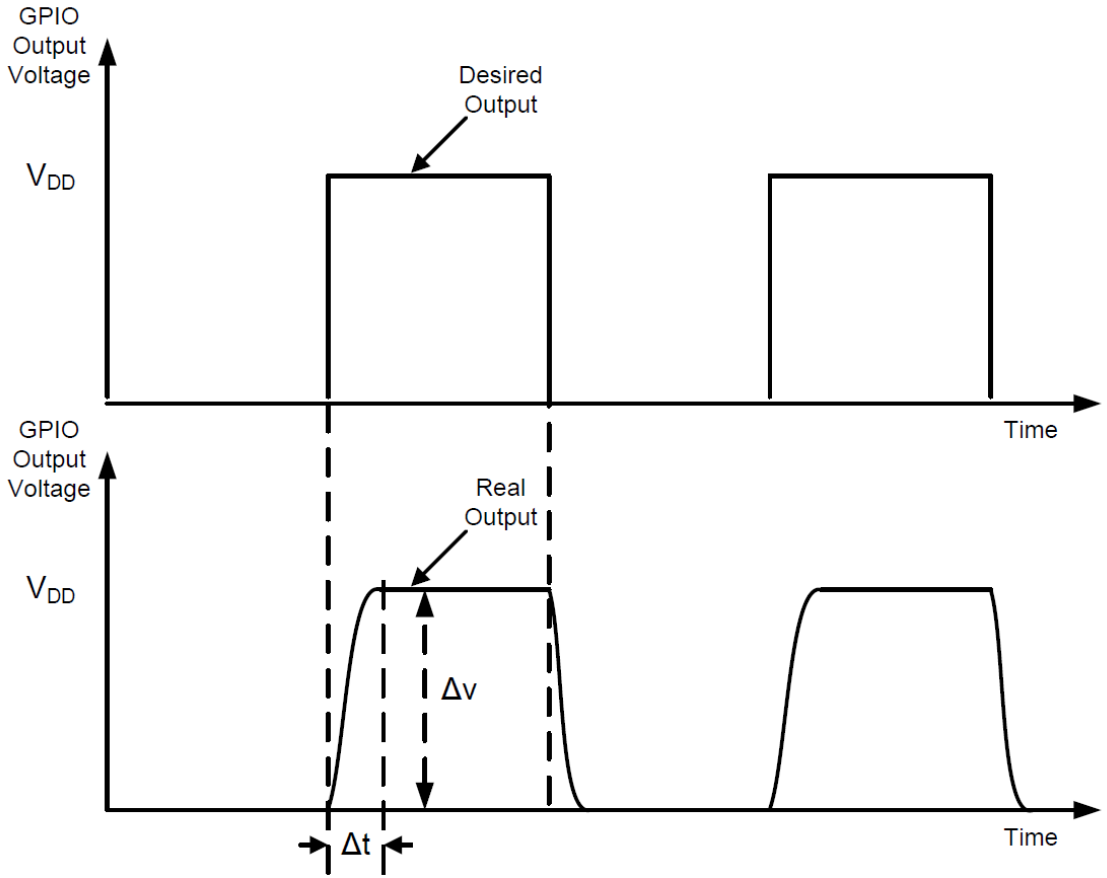
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GPIO: Digital Out

Slew Rate:
Maximum rate of change of the output voltage

$$Slew\ Rate = max\left(\frac{\Delta V}{\Delta t}\right)$$

A high slew rate allows the output to be toggled at a fast speed.



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Case study: LED (PB.2)

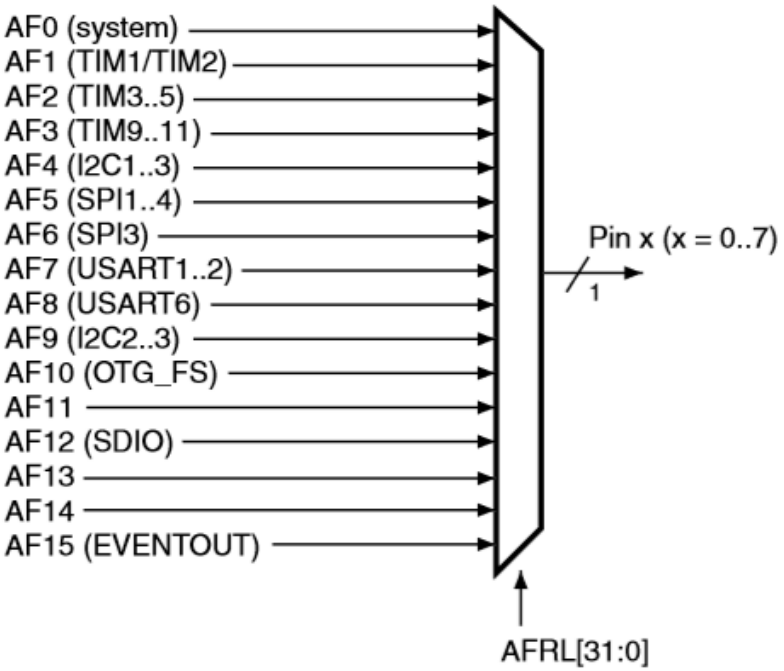
- **Digital Output Configuration Example**
  - LED connected to a pin of GPIO_B

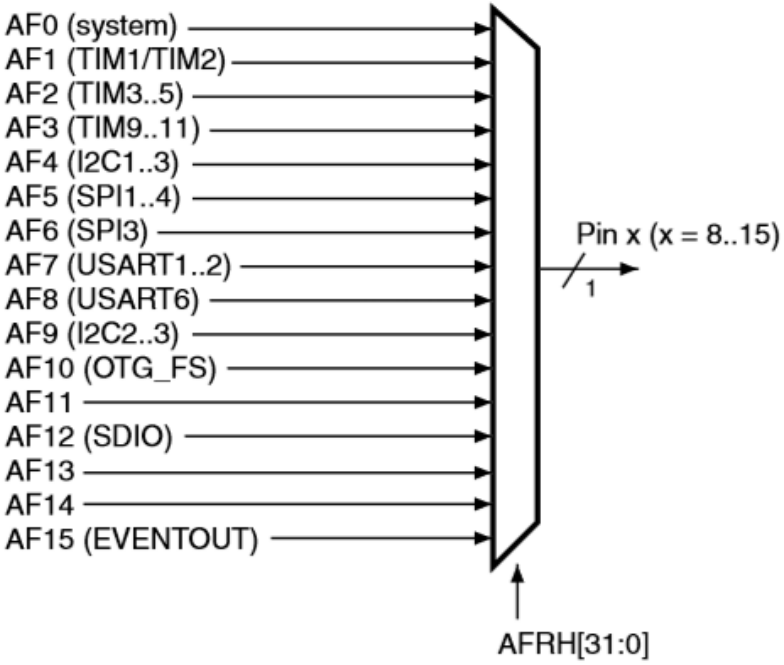
Configuration	Register Name	Value
Enable GPIO B Clock	RCC_AHB2ENR	1=clock enable
Set Pin as Digital Out	GPIO_MODER	01=Digital Out
Select output type	GPIO_OTYPER	0=Push-pull
Select output speed	GPIO_OSPEEDR	00=Low
Select Pull-up/down	GPIO_PUPDR	1=High

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GPIO: Alternate Function

- **Selecting Alternative Function (AF) by**
  - AFRL for Pin x (x=0 to 7)
  - AFRH for Pin x (x=8 to 15)
  - *We will learn more when we cover "timers"*





For pins 8 to 15, the GPIOx_AFRH[31:0] register selects the dedicated alternate function

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Acknowledgement

- Reference & Images are from
  - [Coursera] Coursera "Embedded Software and Hardware Architecture" Univ. Colorado, Boulder
  - [ZHU] Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C, 3rd Ed. Yifeng Zhu
  - [YIU] The Definitive Guide to Arm Cortex-M3 and Cortex-M4 Processors-Newnes (2014), Joseph Yiu (Auth.)
  - [UT] Introduction to the MSP432 Microcontroller: Embedded Systems, Valvano (Free Ebook)